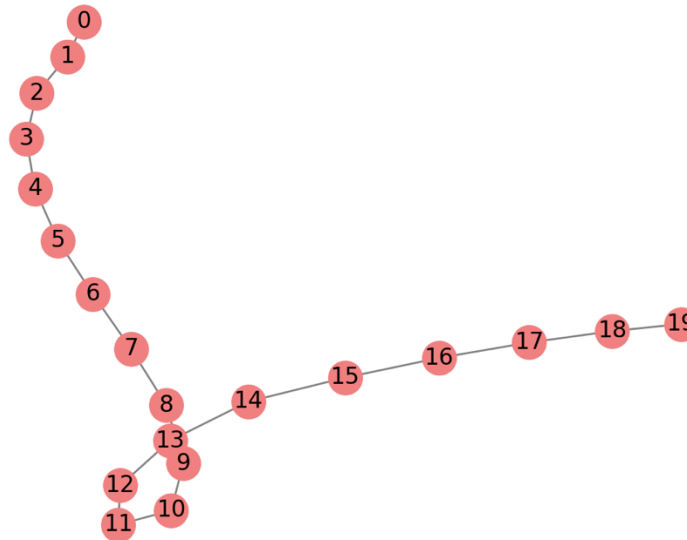
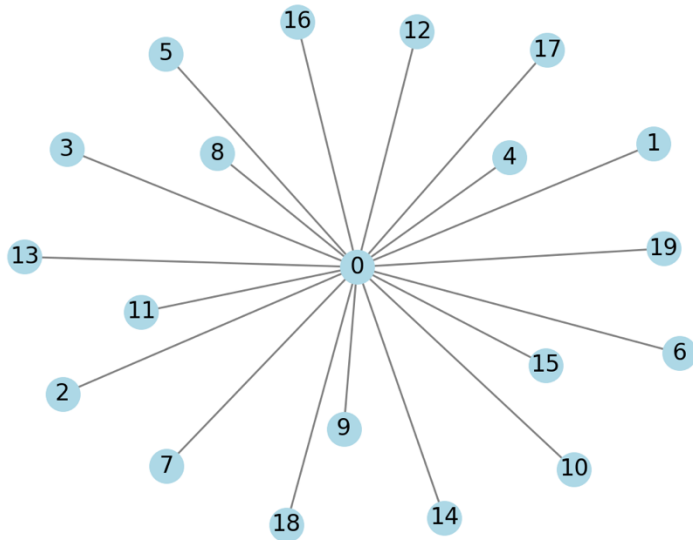


S11: Identify Influencers

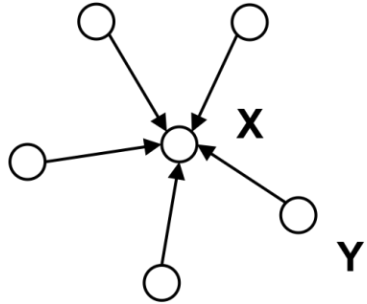
Identify influencers

- You have a marketing campaign
- The goal is to let a post reach more people in very limited time
- The post is not contagious in triggering reposting
- Which network to pick? Which node you want to begin with?

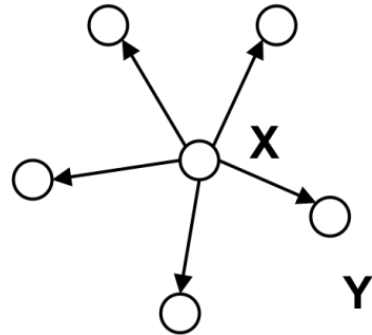


How to identify central nodes?

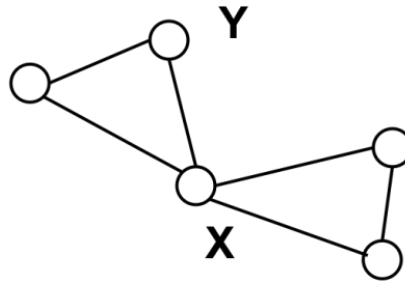
- Which nodes are most “central”?
- Characterize each node’s centrality
- Definition of *central* varies by context/purpose
- In each of the following networks, **X** has higher centrality than **Y** according to a particular measure



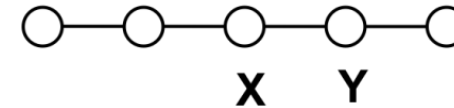
indegree



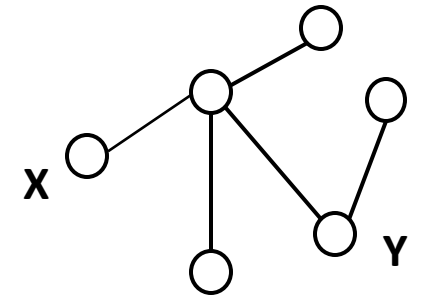
outdegree



betweenness



closeness



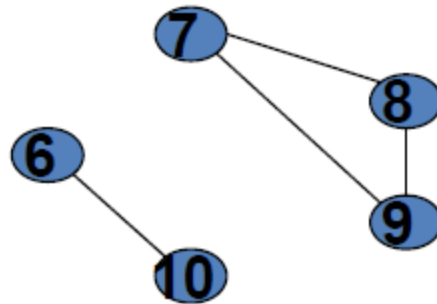
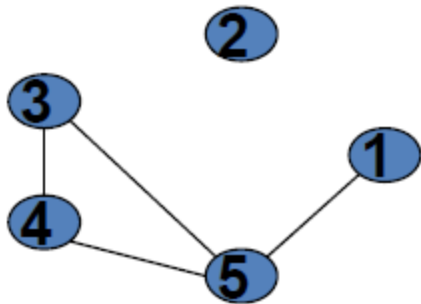
eigenvector

Uses of “centrality”

- **Maximizing Impact:** Influencers can **amplify messages** and drive engagement more effectively than random users.
- **Marketing Efficiency:** Brands can **optimize ad spend** by targeting key individuals who influence many others.
- **Social Movements & Elections:** Political campaigns and social movements can **mobilize supporters** through high-impact individuals.
- **Misinformation Control:** Detecting influential nodes helps **prevent the rapid spread of fake news** or harmful content.
- **Network Resilience:** Organizations can **strengthen networks** by identifying key individuals whose loss would disrupt communication.

Neighborhood and Degree

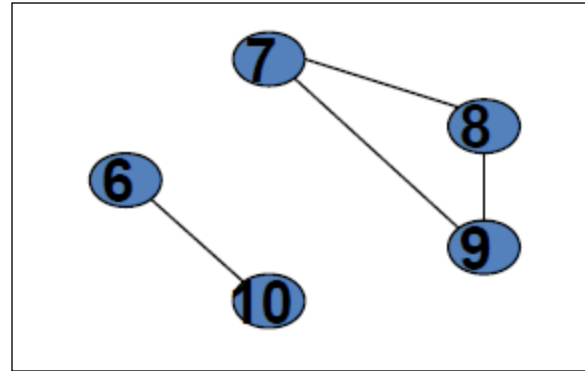
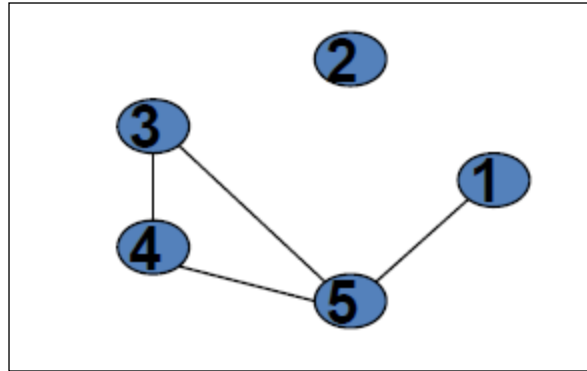
- Neighborhood: directly connected nodes



- Degree: how many neighbors a node has, i.e., total connections.
- In-Degree (Directed Networks): Number of incoming edges (followers on Twitter).
- Out-Degree (Directed Networks): Number of outgoing edges (followings on Twitter).

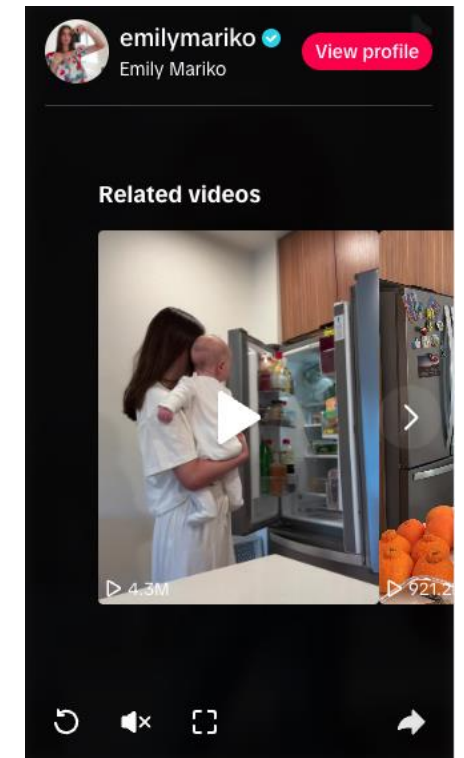
Practice

Which node has the highest degree?



Degree

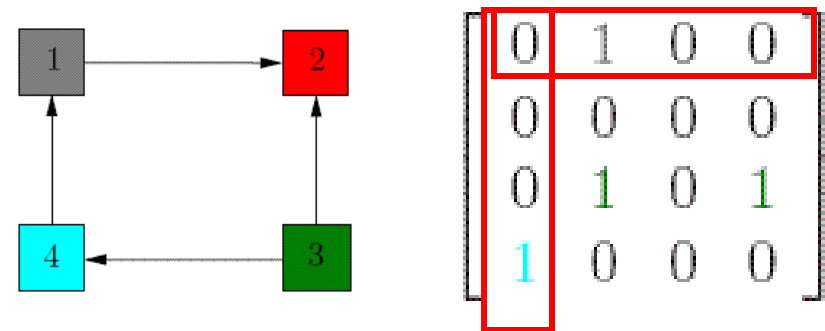
- A **centrality** measure: often used as measure of a node's degree of connectedness and hence also influence and/or popularity
- Useful in assessing which nodes are central with respect to spreading information and influencing others in their immediate 'neighborhood'
- **TikTok influencer pricing**
 - Nano-influencers (1000–10,000 followers): \$5 to \$25 per post
 - Micro-influencers (10,000–100,000 followers): \$25 to \$125 per post
 - Mid-tier influencers (100,000–500,000 followers): \$125 to \$1,200 per post
 - Macro-influencers (500,000–1 million followers): \$1,200 to \$2,500 per post
 - Mega-influencers (1 million+ followers): \$2,500+ per post



Computation

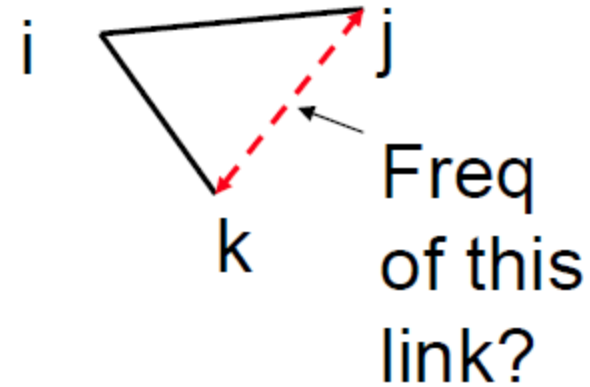
- Degree: Total connections.
- In-Degree (Directed Networks):
Number of incoming edges (followers on Twitter).
- Out-Degree (Directed Networks):
Number of outgoing edges (followings on Twitter).

	v_1	v_2	v_3	v_4	v_5	v_6
v_1	0	1	0	0	0	0
v_2	1	0	1	1	0	0
v_3	0	1	0	1	0	0
v_4	0	1	1	0	1	1
v_5	0	0	0	1	0	0
v_6	0	0	0	1	0	0



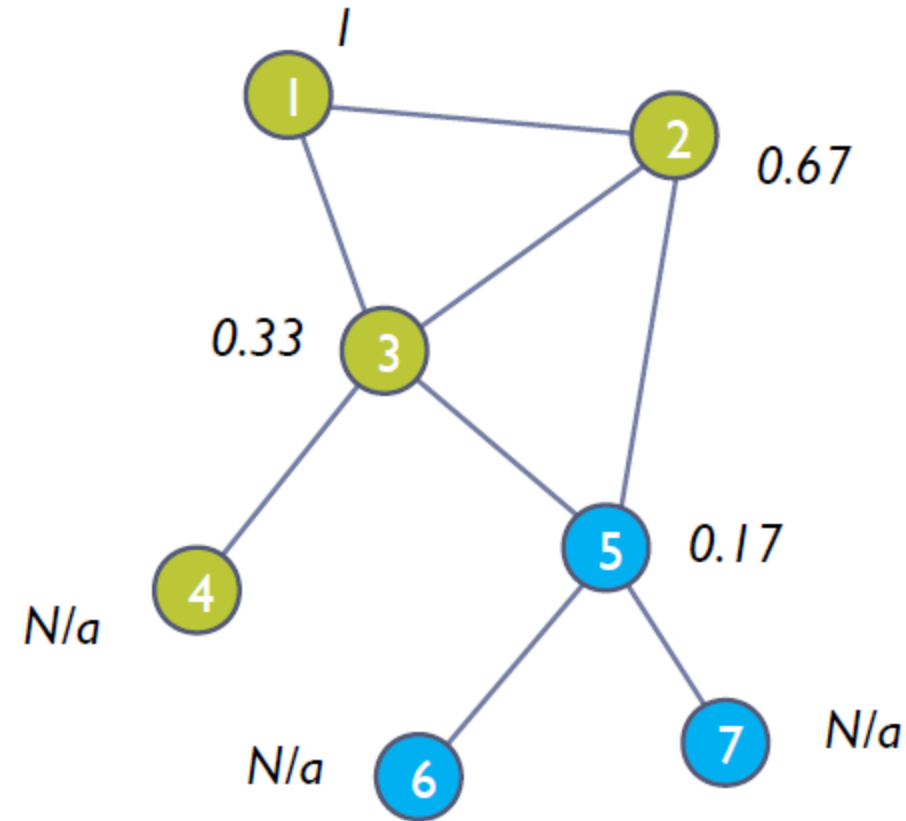
Clustering

- A node's clustering
 - the number of closed triplets in the node's **neighborhood** over the total number of triplets in the **neighborhood**.
- Average clustering (among all nodes)

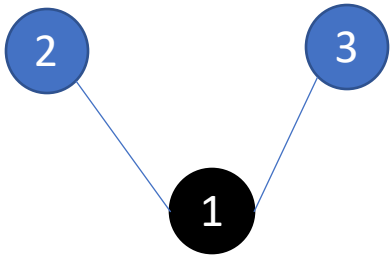


Node clustering

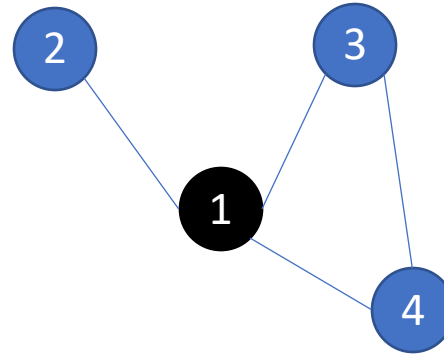
- Node 1 to the right has a value of 1
- Because it is only connected to 2 and 3, and
- These two nodes are also connected
- (i.e. the only triplet in the neighborhood of 1 is closed).
- What about node 2?
- What about node 3?



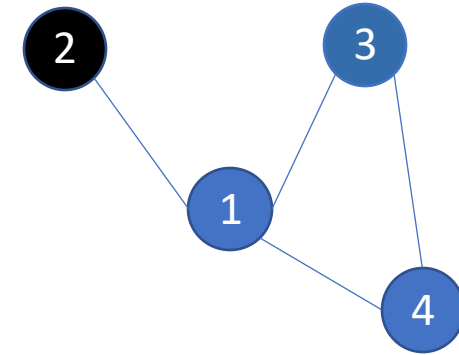
Examples



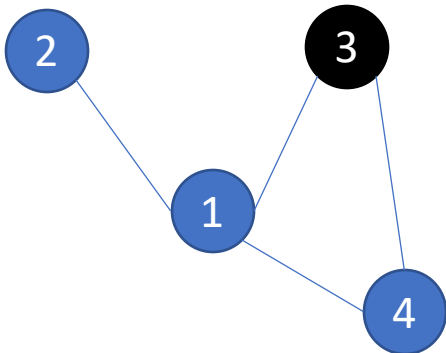
Clustering coefficient for 1?



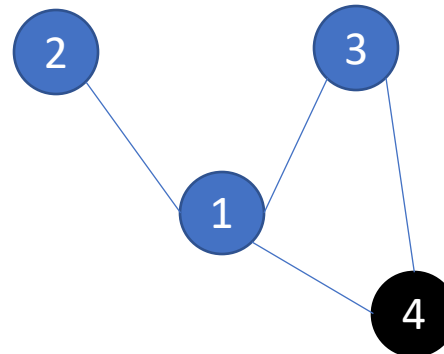
Clustering coefficient for 1 is $1/3$



Clustering coefficient for 2 is 0



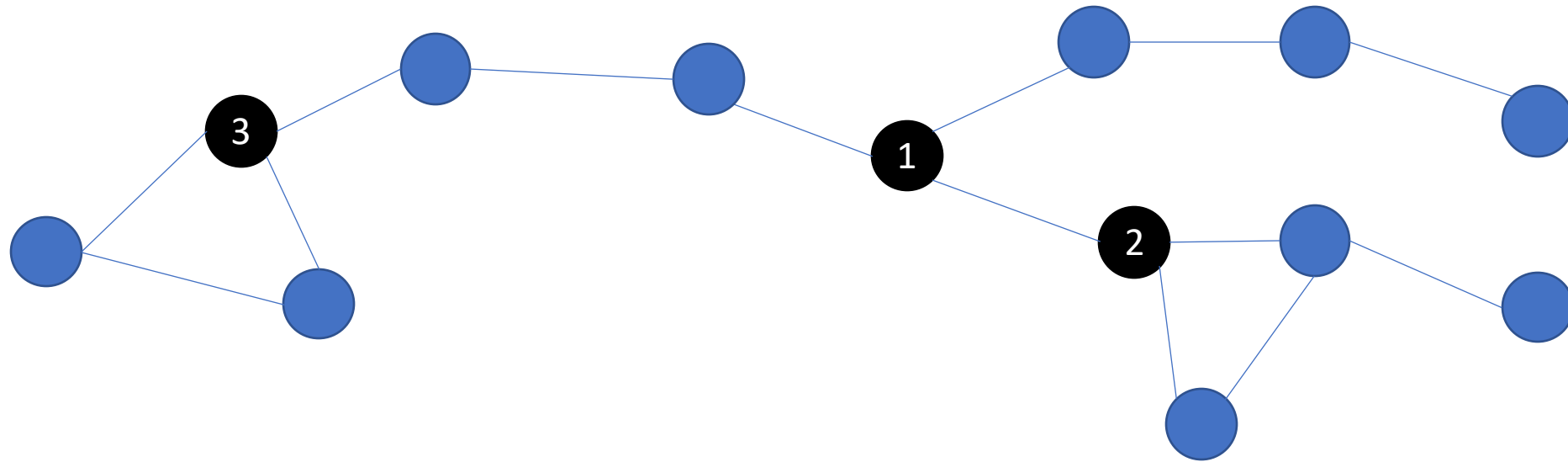
Clustering coefficient for 3 is 1



Clustering coefficient for 4 is 1

Average clustering for the network?

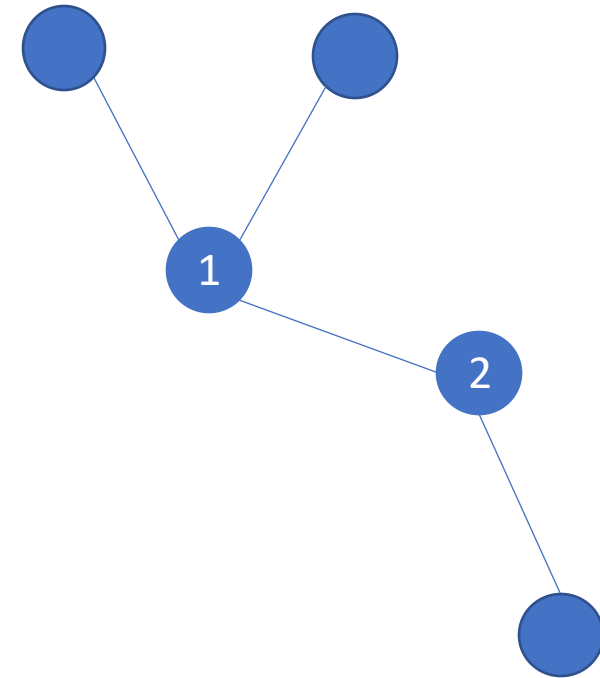
Does degree and clustering capture everything?



- Among nodes 1, 2, and 3, which one has the highest degree?
- Which one has the highest clustering?
- A deadly contagious disease is coming to town, which node would you like to be?
- A juicy gossip is spreading through the network, which node would you like to be? Why?

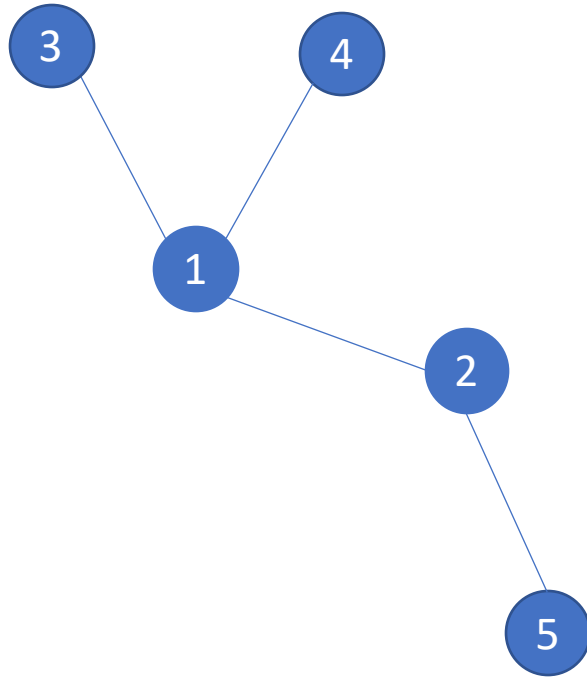
Closeness centrality

- Consider the **geodesic** to every other node!
- Calculate the mean length of the **geodesics** from a node to all other nodes in the network
- How many hops on average it takes to reach every other node)
- Take the reciprocal of the above value $\frac{(n-1)}{\sum_j I(i,j)}$
- Higher values are 'better' (indicate 'more central') like in other measures of centrality
- It is a measure of *reach*, i.e., the *speed* with which information can reach other nodes from a given starting node



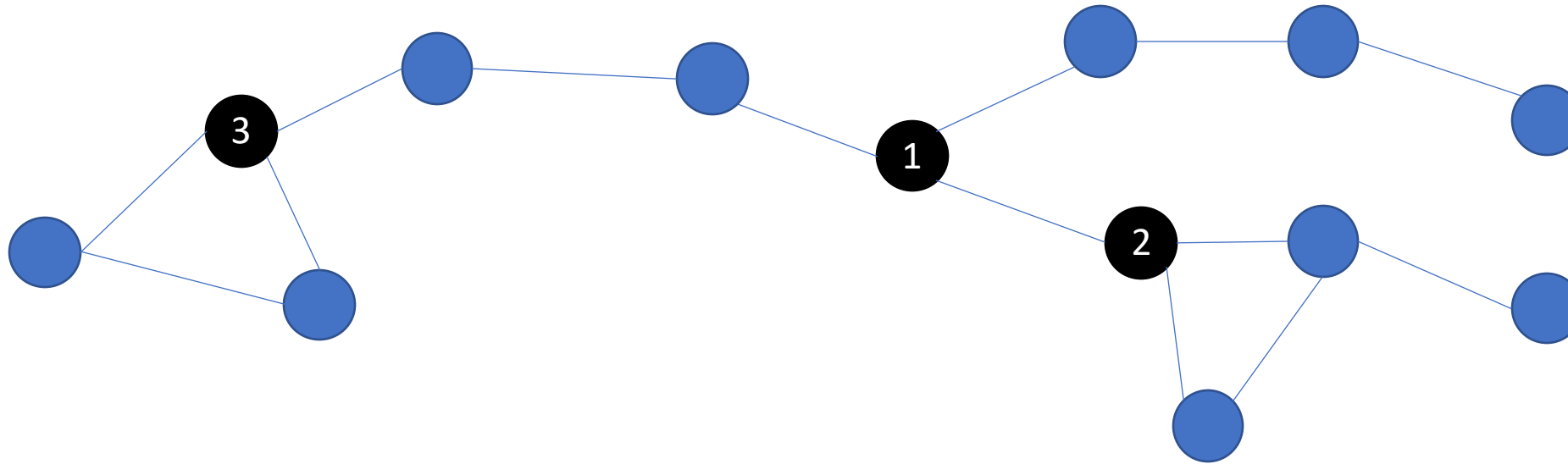
What's the closeness centrality of node 1?

Example



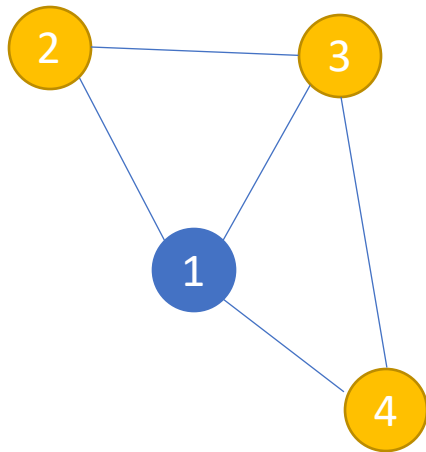
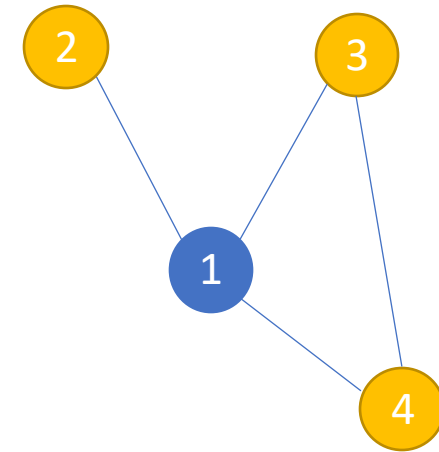
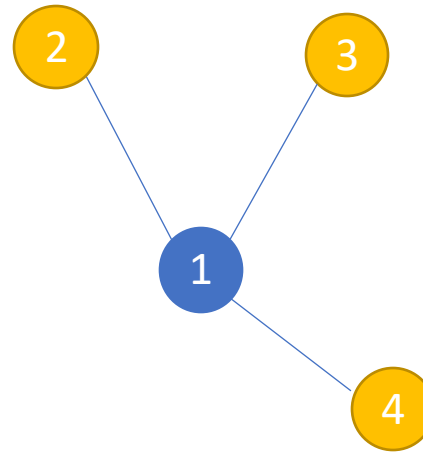
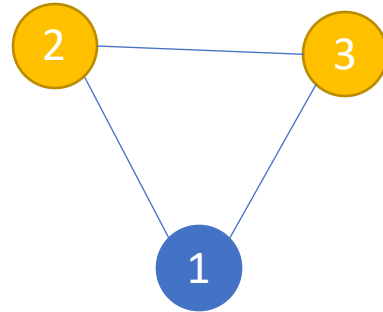
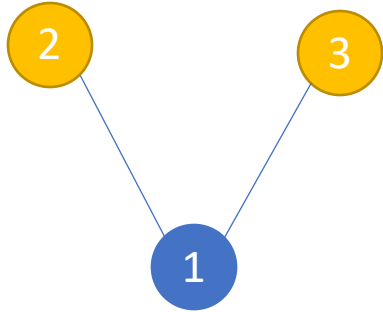
- Closeness centrality of "2":
- Closeness centrality of "3" or "4":
- Closeness centrality of "5":

Betweenness centrality



- **Which nodes are more likely to be in communication paths between other nodes**
- For a given node v , calculate the number of **geodesics** between nodes i and j that pass through v , and divide by all shortest paths between nodes i and j
- Sum the above values for all node pairs i, j
- Sometimes **normalized** such that the highest value is 1 or that the sum of all betweenness centralities in the network is 1

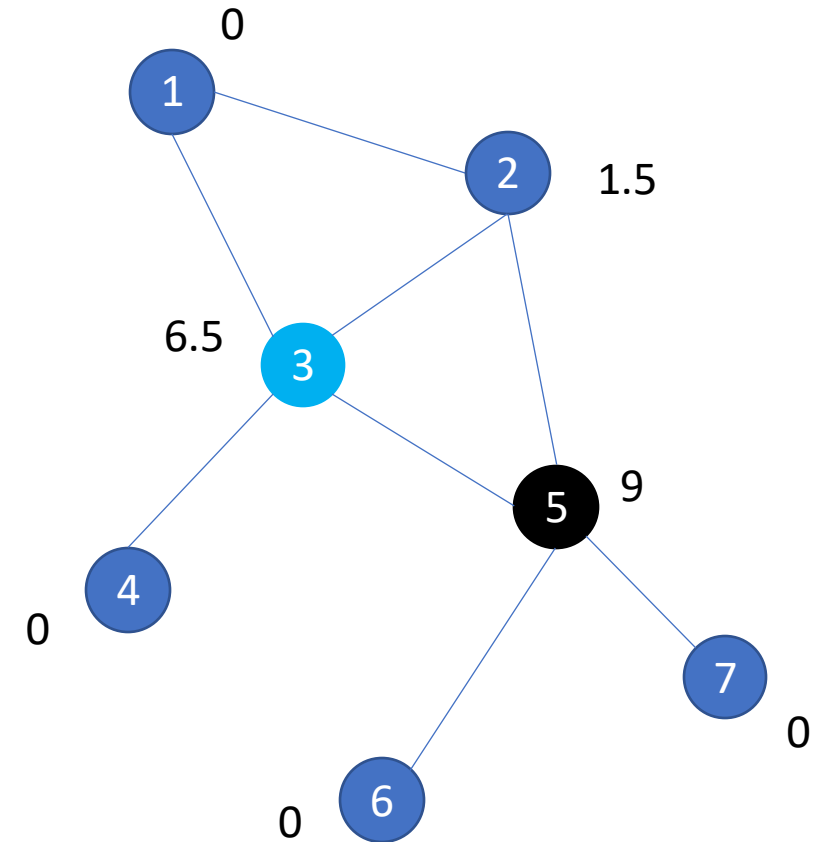
Examples



Use of betweenness centrality

- If my goal is to dissect the network
- How to determine which nodes to cut?
- Useful in determining nodes where the network would break apart

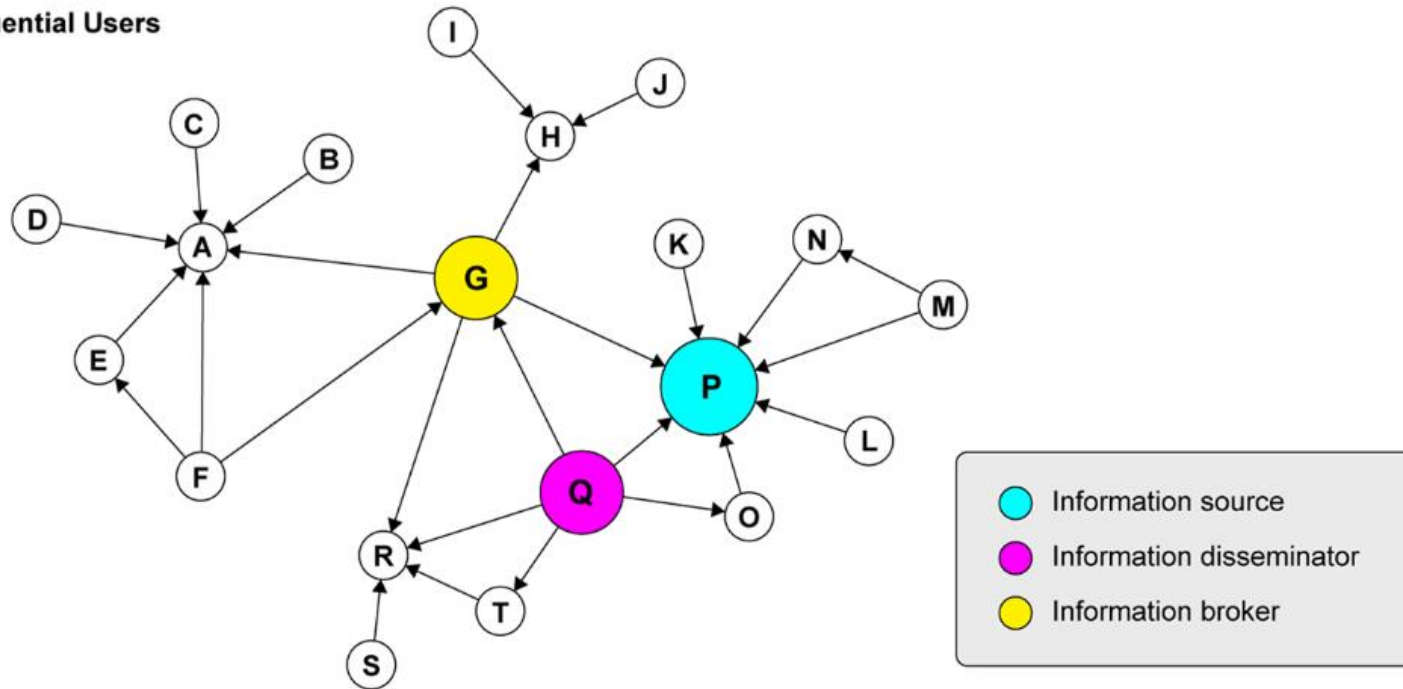
Node 5 has higher betweenness centrality than 3



Source: Prof. Giorgos Cheliotis

Influencer types

C. Influential Users



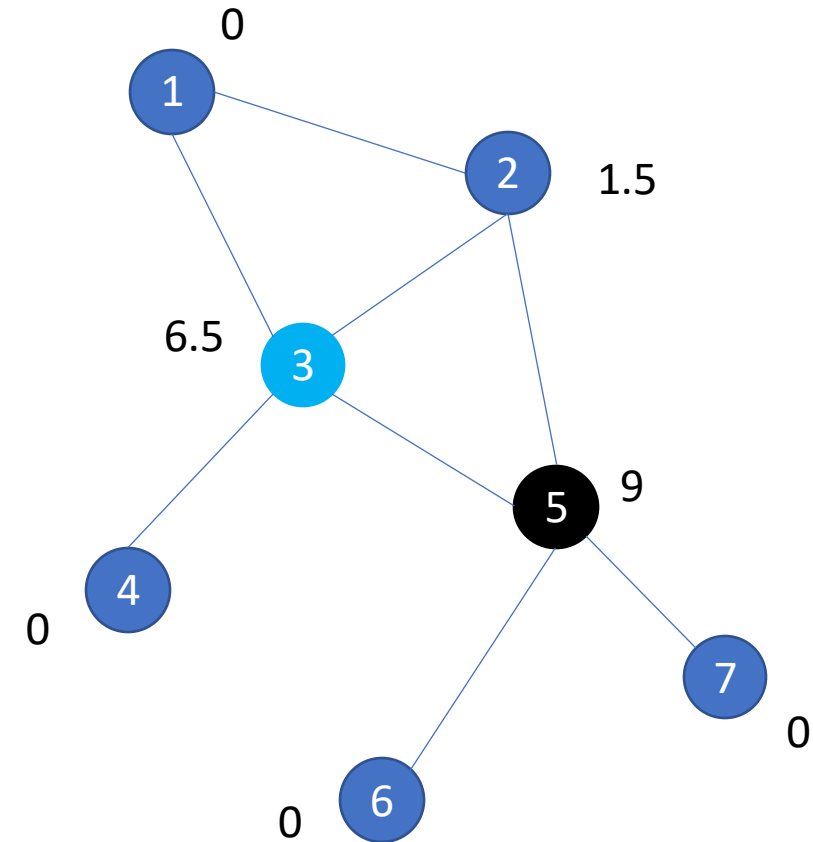
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
IN-DEGREE	6	0	0	0	1	0	2	3	0	0	0	0	1	0	1	7	0	4	0	1
OUT-DEGREE	0	1	1	1	1	3	4	0	1	1	1	1	1	2	1	0	5	0	1	1
BETWEENNESS	0	0	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0

Structural hole

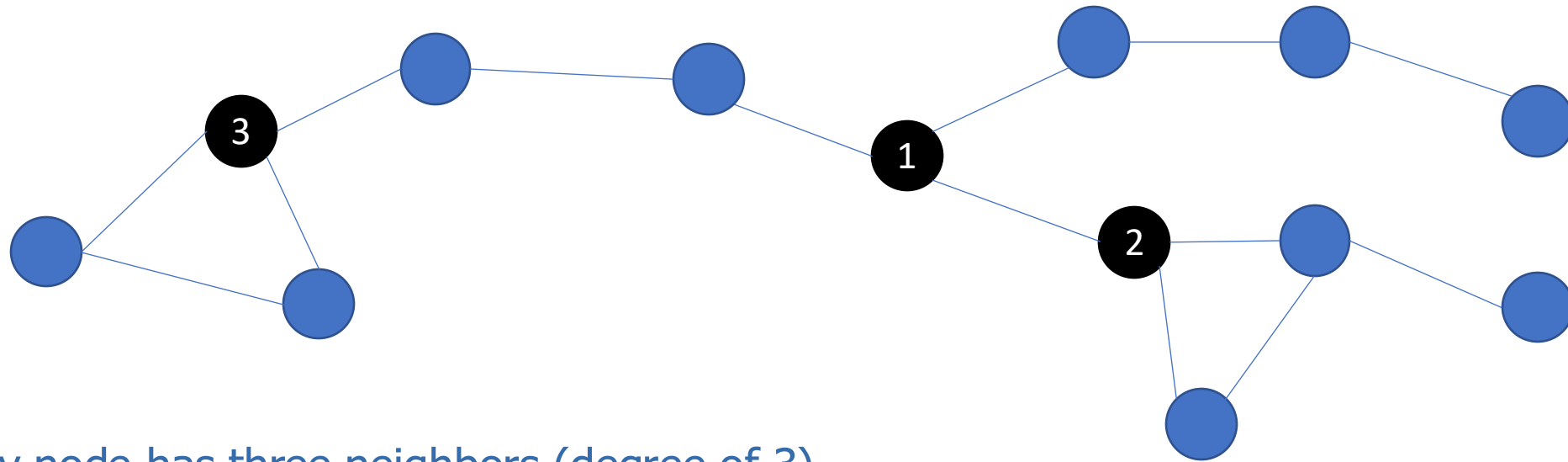
A structural hole refers to **a gap** between two parts of a network that are not directly connected but could be bridged by a single node. A structural hole occurs where a node serves as a bridge between otherwise disconnected groups.

These nodes, **if exist**, would exhibit high **betweenness centrality**. The absence of (a) high betweenness nodes

- Structural holes highlight the absence of connections in a network, emphasizing the potential for a node to act as a **broker** or **mediator** between groups.
- Nodes that bridge structural holes gain **competitive advantage** because they control information or resource flow between disconnected parts of the network.
- Example:
 - A manager who connects two departments that do not communicate directly is filling a structural hole.



Degree with weights (importance of nodes)



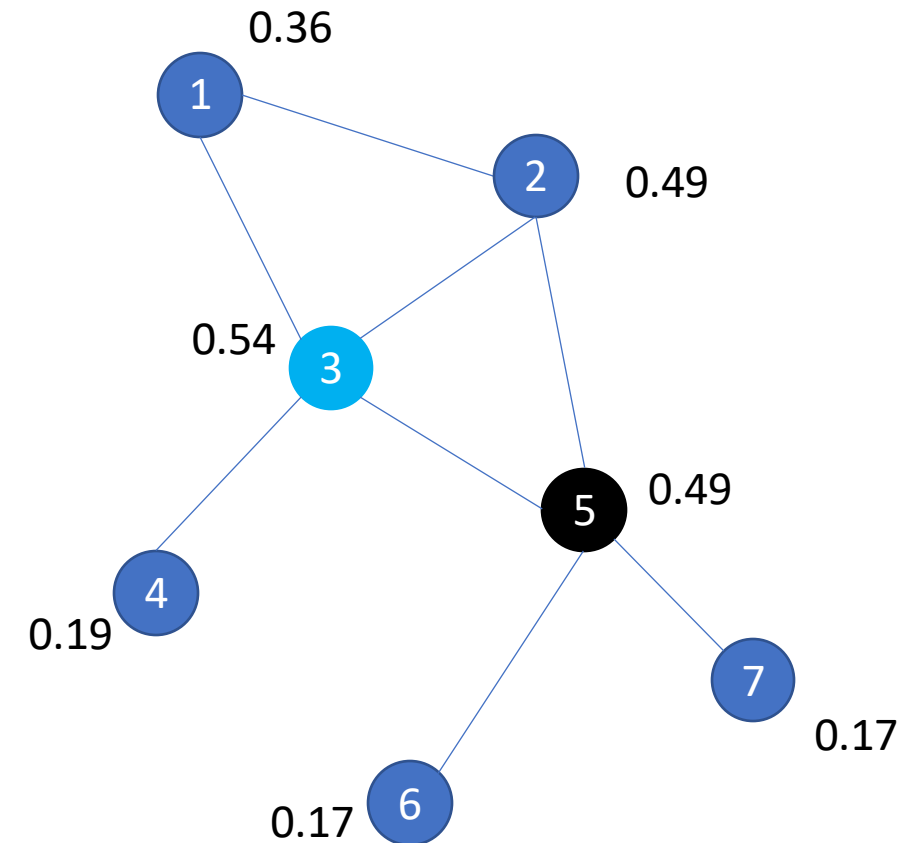
- Every node has three neighbors (degree of 3)
- Whose neighbor are more important?
- The centrality should be proportional to the centralities of neighbors
- In other words, a node with a high centrality is connected to other nodes with high centrality

Eigenvector centrality

- **Finding the principal eigenvector** of the adjacency matrix A

$$x_i = \frac{1}{\lambda} \sum_j A_{ij} x_j$$

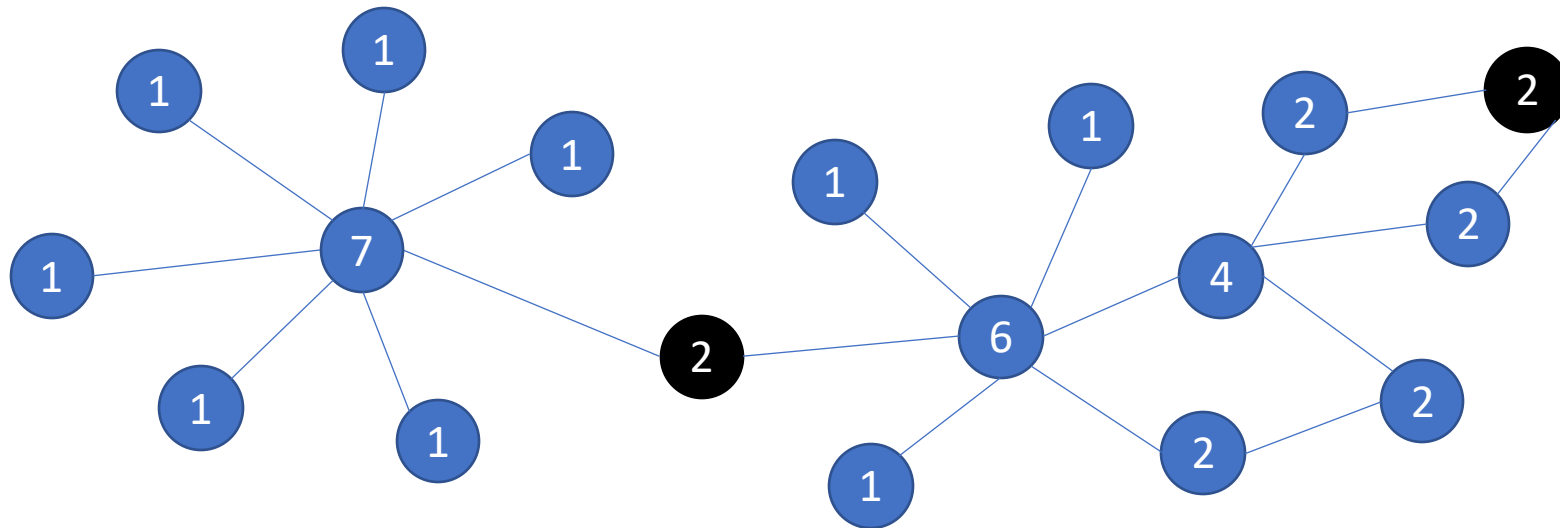
- A node's eigenvector centrality is proportional to the sum of the eigenvector centralities of all nodes directly connected to it
- In other words, a node with a high eigenvector centrality is connected to other nodes with high eigenvector centrality
- Useful in determining who is connected to the most connected nodes



- Node 3 has the highest eigenvector centrality, closely followed by 2 and 5

Example

- Which black node has a higher eigenvector centrality?



Summary of node-level centrality metrics

Degree

How likely is this person to be the most direct route between two people in the network?

Betweenness

How many people can this person reach directly?

Closeness

How well is this person connected to other wellconnected people?

Eigenvector

What fraction of my friends are friends of each other?

Clustering

How fast can this person reach everyone in the network?

Python

- ```
Compute centrality metrics
degree_centrality = nx.degree_centrality(G)
eigenvector_centrality = nx.eigenvector_centrality(G)
betweenness_centrality = nx.betweenness_centrality(G)
closeness_centrality = nx.closeness_centrality(G)
clustering_coefficient = nx.clustering(G)
```
- ```
print(degree_centrality )
print(eigenvector_centrality )
print(betweenness_centrality)
print(closeness_centrality)
print(clustering_coefficient)
```

Python

```
# Find the node with the highest degree
max_degree_node = max(degree centrality, key=degree centrality.get)
max_degree_value = degree centrality[max_degree_node]
print(f"Node with highest degree centrality: {max_degree_node} (Degree Centrality: {max_degree_value:.4f})")
```

```
# Displaying the top 5 nodes with the highest degree centrality
top_5_nodes = sorted(degree centrality.items(), key=lambda x: x[1], reverse=True)[:5]
print("Top 5 Nodes with Highest Degree Centrality:")
for node, value in top_5_nodes:
    print(f"Node {node}: {value:.4f}")
```

```
# Visualizing the network with node size based on degree centrality
node_sizes = [v * 2000 for v in degree centrality.values()]
plt.figure(figsize=(10, 8))
pos = nx.spring_layout(G, seed=42)
nx.draw(G, pos, with_labels=True, node_color="lightblue", edge_color="gray", node_size=node_sizes)
plt.title("Social Network with Node Sizes Proportional to Degree Centrality")
plt.show()
```

MGP 5a

- Your team provides consulting solutions for an election campaign on social media. The campaign will disseminate promotional posts to some users (seeding) and let it evolve within the network.
- Due to the budget constraint, you can only have very limited seeds (like 1 person).
- Your client needs your advices based on different what-if goals.
- In specific, please provide a list of 5 candidates of seeds

```
import networkx as nx
```

```
G1 = nx.read_edgelist("graph1.edgelist", nodetype=int)  
#Please put the file under your pycharm project folder first
```

MGP 5a

- If the post is highly (near a probability of 1) and constantly (across different person) contagious in triggering reposting, and the goal is to reach everyone in shorter time.
- If the post is not contagious in triggering reposting, and the goal is to reach more persons.
- If the post is contagious, and the goal is to first reach the other influential persons.
- If the post is contagious, and the goal is to first reach the well-connected communities (subgroups).
- If the post is not contagious but a fact-checking that can counter other campaigns' post and the goal is to best counter the spreading out of other other campaigns' post.

